

Lab Exercise 2 - Recursion

Lab 2.1: Recursive Calculation of Power of 2

Objective: Develop a recursive solution to calculate n raised to the power of 2 using the given recursive function in C++. Identify and fix any issues in the code.

Instructions:

1. Below is a recursive function written in C++ to calculate n^2 (the power of 2 for a given integer n). The code provided lacks a termination condition. Run the program as given and observe any problems that arise.
2. Modify the code as needed to correct the logic and ensure it terminates correctly.
3. **Marks:** 5

Problem Statement:

- **Original Code:**

```
int powerOfTwo(int n) {  
  
    return n + powerOfTwo(n); // Recursive call  
  
}
```

- **Tasks:**

- Identify the issue with this recursive code and modify it to avoid infinite recursion.
- Add the necessary base case to ensure that the recursion terminates when appropriate.
- Refactor the recursive logic to calculate n^2 correctly.

- **Example Output:**

- `powerOfTwo(3)` should return 9 (since $3^2=9$).
- `powerOfTwo(0)` should return 0.
- `powerOfTwo(1)` should return 1.

4. **Expected Solution Format:**

- Present the corrected code with comments explaining the base case and recursive logic/case.
- Provide a brief explanation of any changes made to address the infinite recursion and correctly compute n^2 .

Lab 2.2: Case Study - Recursive Sum of Integers

Objective: Develop a C++ program using recursion to calculate the sum of all numbers from 1 to a given integer n based on the provided case study. This exercise reinforces understanding of base and recursive cases.

Instructions:

1. Review the case study description and use it to develop a recursive solution in C++.
2. **Marks:** 10

Case Study Description:

- **Scenario:** You are tasked with creating a recursive function that calculates the sum of all integers from 1 up to a given positive integer n . If n is less than or equal to 0, the function should return 0.
 - **Description of Recursive Approach:**
 - **Base Case:** If n is less than or equal to 0, return 0 to terminate recursion.
 - **General Case:** For any positive integer n , return n plus the sum of all numbers from 1 to $n-1$. This recursive case reduces n until it reaches the base case.
3. **Tasks:**
 - Implement the recursive C++ function based on the case study description.
 - Ensure the function properly handles various input values, including edge cases like $n = 0$ and negative numbers.
 - Write test cases to verify the function's output.
 - Not permitted to use the `pow` function in C++ to calculate powers of numbers
 4. **Expected Solution Format:**
 - Present the complete C++ code with comments explaining the base case and recursive step.
 - Provide sample output for test cases, such as `sumUpTo(5)` which should return 15 ($1 + 2 + 3 + 4 + 5 = 15$).

Notice on Generative AI Tool Usage

Students are encouraged to develop their solutions to reinforce learning and problem-solving skills. Submissions that are identified **as being primarily generated by AI tools** will receive a reduced grade of 7 out of 15, as such work may not fully demonstrate individual/team understanding and effort.

Individual Submission (Page 3 and 4 only)

Group 9:

Lab 2.1 Answer

```
#include <string>

using namespace std;

int powerOfTwo(int n) {
    if (n <= 0) {
        return 0;
    } else if (n == 1) {
        return 1;
    } else {
        return n + powerOfTwo(n-1) + (n - 1);
    }
}

int main() {
    int v = powerOfTwo(2);
    int w = powerOfTwo(0);
    int x = powerOfTwo(4);
    int y = powerOfTwo(1);
    int z = powerOfTwo(8);

    cout << v << endl;
    cout << w << endl;
```

```
    cout << x << endl;

    cout << y << endl;

    cout << z << endl;

}
```

Lab 2.2 Answer

```
#include <iostream>

#include <string>

using namespace std;

int sumUpTo(int n) {
    if (n <= 0) {
        return 0;
    } else if (n == 1) {
        return 1;
    } else {
        return n + sumUpTo(n-1);
    }
}
```

```

    }

}

int main() {

    int c;

    c = sumUpTo(5);

    cout << c << endl;

    return 0;

}

```

Total Score: /15 (Each student rates themselves and each of their peers individually)
 Submission: Each student submits an individual assessment of their own contributions and those of their group members based on contributions and teamwork.

Evaluator/Criteria	Technical Skills(/4)	Idea Contribution(/4)	Engagement & Seriousness (/4)	Collaboration & Team Spirit (/3)	Total(/15)
Nurul Adriana Binti Kamal Jefri (Self-evaluation)	4	4	4	3	15

DHESHIEGHAN A/L SARAVANA MOORTHY	4	4	4	3	15
TEH RU QIAN	4	4	4	3	15
LAU YAN KAI	4	4	4	3	15